



Cambridge International AS & A Level

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NAME
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FURTHER MATHEMATICS

9231/32

Paper 3 Further Mechanics

October/November 2024

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

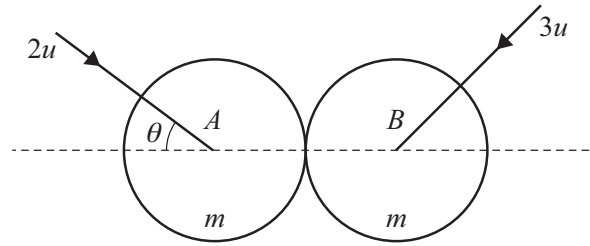
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- Find the extension of the string. [3]

[illegible]





The diagram shows two identical smooth uniform spheres A and B of equal radii and each of mass m . The two spheres are moving on a smooth horizontal surface when they collide with speeds $2u$ and $3u$ respectively. Immediately before the collision, A 's direction of motion makes an angle θ with the line of centres and B 's direction of motion is perpendicular to that of A . After the collision, B moves perpendicular to the line of centres. The coefficient of restitution between the spheres is $\frac{1}{3}$.

- (a)** Find the value of $\tan \theta$.

[3]

[illegible]



(b) Find the total loss of kinetic energy as a result of the collision. [2]

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(c) Find, in degrees, the angle through which the direction of motion of A is deflected as a result of the collision. [2]

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- (b) Find, in terms of W , the magnitude of the frictional force between the rod and the wall. [2]

[illegible]

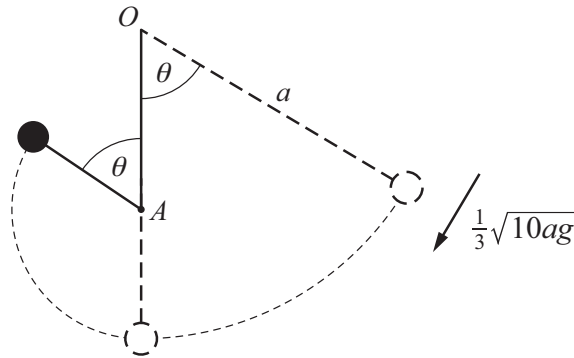
- (a) Use the equation of the trajectory to show that $u^2 = 25ag$. [2]

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(b) Express V^2 in the form kag , where k is a rational number. [6]

[illegible]



A particle P of mass m is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point O . The particle P is held with the string taut and the string makes an angle θ with the downward vertical through O . The particle P is projected at right angles to the string with speed $\frac{1}{3}\sqrt{10ag}$ and begins to move downwards along a circular path. When the string is vertical, it strikes a small smooth peg at the point A which is vertically below O . The circular path and the point A are in the same vertical plane. After the string strikes the peg, the particle P begins to move in a vertical circle with centre A . When the string makes an angle θ with the upward vertical through A the string becomes slack (see diagram). The distance of A below O is $\frac{5}{9}a$.

- (a) Find the value of $\cos \theta$.

[6]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

- (a) Find an expression for v in terms of t . [6]

[illegible]



(b) Find an expression for v^2 in terms of x .

[5]

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[illegible]

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